편도함수

연습문제 풀이

13

①

연습문제 13.1

1. 다음 함수의 정의역을 구하여라.

(1) $f(x,y) = \sqrt{x} + \sqrt{y}$

$\sqrt{x}$, $\sqrt{y}$ 는 0보다 커야한다.

$\therefore$ $\{(x,y) : x \geq 0$ 그리고 $y \geq 0\}$

(2) $f(x,y) = \ln \dfrac{x^2 + y^2}{(x^2 - y^2)^2}$

분모는 0이 될 수 없다.

$\therefore \{(x,y) : x \neq y,\ x \neq -y$ 인 모든 실수$\}$

연습문제 13.2

1. 다음 극한값을 구하여라.

(1) $\lim_{(x,y)\to(1,0)} \frac{x^2-xy+1}{x^2+y^2}$

$x=1, y=1$으로 접근하므로

$$\lim_{(x,y)\to(1,0)} \frac{1^2-(1)(0)+1}{1^2+0^2}=2$$

$\therefore 2$

연습문제 13.3

1. 다음 함수의 1계도함수 및 2계도함수를 구하여라.

(1) $f(x,y) = 2x + 3x^2y^4$

1계도함수

$f_x(x,y) = 2+3y^4(2x) = 2+6xy^4$

$f_y(x,y) = 0+3x^2(4y^3) = 12x^2y^3$

2계도함수

$f_{xx}(x,y) = 0+6y^4(1) = 6y^4$

$f_{xy}(x,y) = 0+6x(4y^3) = 24xy^3$

$f_{yy}(x,y) = 12x^2(3y^2) = 36x^2y^2$

$f_{yx}(x,y) = 12y^3(2x) = 24xy^3 = f_{xy}(x,y)$

(2) $f(x,y,z) = x(\cos y)e^z$

1계도함수

$f_x(x,y,z) = (\cos y)e^z$

$f_y(x,y,z) = xe^z(-\sin y) = -x(\sin y)e^z$

$f_z(x,y,z) = x(\cos y)e^z$

2계도함수

$f_{xx}(x,y,z) = 0$

$f_{xy}(x,y,z) = e^z(-\sin y) = -(\sin y)e^z = f_{yx}(x,y,z)$

$f_{xz}(x,y,z) = (\cos y)e^z = f_{zx}(x,y,z)$

$f_{yy}(x,y,z) = -xe^z(\cos y) = -x(\cos y)e^z$

$f_{yz}(x,y,z) = -x(\sin y)e^z = f_{zy}(x,y,z)$

$f_{zz}(x,y,z) = x(\cos y)e^z$

2. 주어진 점에서 f의 1계편도함수를 구하여라.

(1) $f(x,y)=\sqrt{4x^2+y^2}$, $(2,-3)$

$$f_x(x,y)=\frac{\partial}{\partial x}(4x^2+y^2)^{\frac{1}{2}}=\frac{1}{2}(4x^2+y^2)^{-\frac{1}{2}}(8x)$$

$$=\frac{4x}{\sqrt{4x^2+y^2}}$$

$$f_x(2,-3)=\frac{4(2)}{\sqrt{4(2)^2+(-3)^2}}=\frac{8}{5}$$

$$f_y(x,y)=\frac{\partial}{\partial y}(4x^2+y^2)^{\frac{1}{2}}=\frac{1}{2}(4x^2+y^2)^{-\frac{1}{2}}(2y)$$

$$=\frac{y}{\sqrt{4x^2+y^2}}$$

$$f_y(2,-3)=\frac{(-3)}{\sqrt{4(2)^2+(-3)^2}}=-\frac{3}{5}$$

(2) $f(x,y,z) = xy^2 \sin z, \ (-1,2,0)$

$$f_x(x,y,z) = (1)y^2 \sin z = y^2 \sin z$$
$$f_x(-1,2,0) = (2)^2 \sin(0) = 0$$

$$f_y(x,y,z) = x(\sin z)(2y) = 2xy(\sin z)$$
$$f_y(-1,2,0) = 2(-1)(2)\sin(0) = 0$$

$$f_z(x,y,z) = xy^2 \cos z$$
$$f_z(-1,2,0) = (-1)(2)^2 \cos(0) = -4$$

4. f의 2계도함수를 구하여라.

(1) $f(x,y) = 3x^2 - \sqrt{2}\,xy^2 + y^5 - 2$

$$f_x(x,y) = 6x - \sqrt{2}\,y^2$$
$$f_y(x,y) = -\sqrt{2}\,x(2y) + 5y^4 = -2\sqrt{2}\,xy + 5y^4$$

$$f_{xx}(x,y) = 6$$
$$f_{xy}(x,y) = -\sqrt{2}\,(2y) = -2\sqrt{2}\,y = f_{yx}(x,y)$$
$$f_{yy}(x,y) = -2\sqrt{2}\,x + 20y^3$$

(4) $f(x,y,z) = z\cos xy$

$$f_x(x,y,z) = z(-\sin xy)(1y) = -zy\sin xy$$
$$f_y(x,y,z) = z(-\sin xy)(1x) = -zx\sin xy$$
$$f_z(x,y,z) = \cos xy$$

$$f_{xx}(x,y,z) = -zy(\cos xy)(1y) = -zy^2\cos xy$$

$$\begin{aligned} f_{xy}(x,y,z) &= -z(\sin xy) + (-zy)\{(\cos xy)(x)\} \\ &= -z(\sin xy) - xyz(\cos xy) = f_{yx}(x,y,z) \end{aligned}$$

$$f_{xz}(x,y,z) = -y\sin xy = f_{zx}(x,y,z)$$

$$f_{yy}(x,y,z) = -zx(\cos xy)(x) = -zx^2\cos xy$$
$$f_{yz}(x,y,z) = -x\sin xy = f_{zy}$$
$$f_{zz}(x,y,z) = 0$$

연습문제 13.4

1. $\dfrac{du}{dt}$를 구하여라.

(1) $u = \ln(3x^2 + y^2),\ x = e^{2t},\ y = t^{1/3}$

$$\frac{du}{dt} = \frac{du}{dx}\frac{dx}{dt} + \frac{du}{dy}\frac{dy}{dt}$$

$$= \left\{\frac{1}{3x^2+y^2}(6x)\right\}(2e^{2t}) + \left\{\frac{1}{3x^2+y^2}(2y)\right\}\left(\frac{1}{3}t^{-\frac{2}{3}}\right)$$

$$= \frac{12xe^{2t}}{3x^2+y^2} + \frac{1}{3}\frac{(2y)(t^{-\frac{2}{3}})}{3x^2+y^2}$$

$$= \frac{12(e^{2t})e^{2t}}{3(e^{2t})^2+(t^{1/3})^2} + \frac{1}{3}\frac{(2t^{1/3})}{3(e^{2t})^2+(t^{1/3})^2}t^{-\frac{2}{3}}$$

$$= \frac{12e^{4t} + \frac{2}{3}t^{-\frac{1}{3}}}{3e^{4t} + t^{\frac{2}{3}}}$$